Methods Supplement

From Hairballs to Hypotheses: A Survey of AI-Assisted Visual Analytics in the Life Sciences

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This supplement accompanies the CGF revision of the manuscript named above. It documents record identity, provisional classification, selection and access, assessment authority, and the per-system evidence underlying Figure 3. The earlier reproducibility package is archived at `https://doi.org/10.5281/zenodo.23084434`. The reserved DOI above identifies the revised methodology supplement and its supporting files. DOI reservation does not indicate that the revised deposit has been published.

1 Deduplication and Related Reports

Deduplication consolidated 187,446 source occurrences into 140,959 bibliographic records. A *source occurrence* is an entry returned by a provider or imported from another source; several occurrences may refer to the same publication. A *canonical record* is the representative bibliographic entry retained after applying the identity rules.

Matching used normalized DOI keys where available and exact normalized-title keys otherwise. Records with and without DOIs were kept separate when title agreement alone did not establish their identity. arXiv identity adjudications were incorporated into the final total. Source occurrences and identity decisions were retained so that canonical records could be traced to their acquisition routes.

Fifty related-version links connect publications such as preprints and published articles. These links preserve relationships between reports that remain separate bibliographic records. Grouping reports about the same software system occurs during full-text assessment, as described in the main paper.

1.1 Retrieval and Screening Accounting

The retrieval cutoff was 20 September 2026, without a lower publication-date limit. The local arXiv snapshot contained 3,164,528 records and yielded 958 distinct query-family matches; observed metadata updates extended through 11 September 2026. Its coverage is distinct from the cutoff for the other retrieval routes. Ehlers-associated working materials, Joos-associated companion material, and the reconstruction of Fonnet and Prié's bibliography contributed 816,

146, and 207 source occurrences, respectively. These entered the same screening workflow as database results; source inventories distinguish confirmed survey members, unconfirmed entries, and background references.

Of 140,959 deduplicated records, 140,810 received valid screening outcomes: 9,505 Advance, 83,258 Defer, and 48,047 Excluded. The other 149 comprised 110 reserved pilot/calibration records, 35 technical failures, and four ambiguous requests that were not resubmitted. These states are not scientific exclusions. Of the excluded records, 10,460 also carried a background-literature flag indicating potential contextual or discovery value; these flags are a subset of exclusions, not additional records.

2 Provisional Candidate Classification

The task groups, assistance modes, and visualization modalities are defined in Section 3 of the main paper. Each label was assigned a Present, Absent, or Uncertain state, and multiple labels could coexist. Classification used evidence for the specific category: for example, unspecified display information yielded uncertainty about modality. The same category definitions were applied to all 9,505 advanced records using their supplied titles and abstracts.

These provisional classifications organized the candidate pool and guided selection for full-text assessment. Table S1 summarizes assistance–modality co-occurrences among candidate records. A record contributes to a cell when both labels are Present; cells overlap. In the export, Present and Uncertain labels are listed explicitly, with omission denoting Absent under the classification contract. Desktop/Planar, for example, was Present in 2,912 records and Uncertain in 4,072.

The table describes labels supported by bibliographic metadata. Establishing that a particular assistance mechanism operates within a particular display environment requires assessment of the system description, which underlies Figure 3 of the main paper.

2.1 Classification Contract and Interpretation

Present denotes positive title/abstract evidence for a label; Absent denotes a negative category assignment under the coding contract; Uncertain records insufficient or ambiguous evidence. Absent is therefore not a full-report finding that a capability is missing. Assistance, modality, and task labels may coexist. Their definitions are those in Section 3 of the main paper, including substantive computation for Algorithmic assistance, a signal-linked computational intervention for Adaptive assistance, interpreted language for Conversational assistance, and computational use of spatial or embodied context for Immersive assistance. Display size or a headset alone does not supply the latter evidence.

The candidate table counts records with both assistance and modality marked Present, without asserting that the mechanism operates in that environment. Framework placements instead require report-level evidence for that relationship. Historical export fields and labels remain unchanged; the manuscript's eligibility criteria are E1–E5. The archived prompts, protocols, and screening settings preserve the execution specifications rather than a retrospective reconstruction of them.

3 Purpose-Specific Selection Rules

The 384 selected report identities represent six sets and 381 systems, not 384 completed assessments or one uniform sample. Of the report identities, 376 link exactly to the terminal 9,505-record pool. Three additional systems entered through coauthor-supplied reports and five through bounded early map expansion. A distinct prior-survey-only full-report selection route is not established by the cumulative crosswalk.

Table S1: Machine-coded candidate co-occurrences among 9,505 records. Both labels must be PRESENT; cells overlap and are not additive. These are discovery counts, not verified mechanism placements. D: Desktop/Planar; L: Large Display; A: AR/MR.

Assistance	D	L	VR	A	CAVE
Algorithmic	2,837	37	511	399	11
Adaptive	806	12	185	190	2
Conversational	86	1	15	22	0
Immersive	112	20	428	346	8

3.1 Initial 20 entries

Purposive anchors from the lead author's prior reading and coauthor-supplied reports were combined with assessed coauthor reports to establish the mechanism map. The candidate export resolved identities of already chosen examples, rather than ranking them. The lead author supplied the twelve anchor nominations; a Codex agent nominated the replacement and seven additions from coauthor-source evidence. The initial 20 was a resulting set, not a documented numerical cap. The twelve anchors were not ordered or capped in advance; no formal tie-breaking rule was recorded for this purposive set.

3.2 Bounded 18 entries

A frozen queue excluded previously represented systems and prioritized reusable lawful local reports, machine-coded candidates for empty cells, sparse Large Display/AR/MR/CAVE cases with explicit mechanism leads, and bounded contrast cases. Per-entry rationales and queue order are retained. The lead author specified the priorities and a maximum of 40; a Codex agent authored the exact 18-entry queue. The resulting queue contained 18 nominations against those priorities within the 40-entry bound; it was not a fixed-size sample. Within-priority tie rules were not recorded.

3.3 Expanded 50 entries

Software ranked candidates after excluding prior identities. Scores reflected rare modalities, non-desktop Adaptive/Conversational labels, mechanism/evaluation terms, contrast terms, and access indicators; ordering used descending score then lowercase title. The final 50 were a separately specified list with per-entry rationales, not the ranked top 50. Software validated and materialized that list within a maximum of 60. A Codex agent curated the final nominations under the lead author's maximum of 60. The resulting list contained 50 nominations against the stated priorities within the 60-entry bound; it was not a fixed-size sample or the top 50 of the ranked export. Selection among similarly suitable records was not governed by a recorded tie-breaking rule.

3.4 Deterministic 200 entries

Software validated the 9,505-record input and excluded previously considered canonical IDs, exact titles and DOIs, and subsequent exact DOI/title duplicates. Scores combined sparse-cell coverage, workflow, life-science, interaction, computation, evaluation/mechanism terms, abstracts, likely open access, assistance/modality bonuses, and contrast indicators. The implemented ordering was high-precision signal, rationale class, likely-open-access status, descending score, then canonical identifier. The lead author authorized the 200-entry bound and limit of 140 completed new assessments. The first 200 were frozen: 1–140 were primary and 141–200 ordered reserves. Retrieval continued through queue exhaustion; the maximum of 140 concerned completed new

assessments rather than selected report identities. Only 60 of the 200 report entries had accessible full text at the September cutoff; grouping related reports yielded 57 completed system assessments. Access, rather than the assessment ceiling, therefore limited completion. The original audit JSON's textual sort description omits the high-precision and open-access strata. A dated correction overlay, `queue_selection_audit_correction_20261009.json` in the same staging directory, records the implemented order and binds it to the executable script. The original audit remains unchanged; no selection was rerun and no queue membership, order, or assessment changed.

3.5 Sparse-cell 24 entries

An upstream 25-entry shortlist from an 8,832-record partial export allowed up to three candidates per sparse cell, favored abstracts and DOIs, and excluded prior exact IDs/titles. Software rebound the entries to the terminal 9,505 export and checked canonical IDs, DOI, normalized title, and related versions. Removing AREA 2013 as a related report of previously considered AREA 2012 yielded 24 entries in preserved priority order, within a maximum of 30. The lead author supplied the provisional 25-entry list and authorized the maximum of 30; software performed reconciliation. The supplied shortlist, rather than a fresh software ranking, determined membership; reconciliation determined the final count of 24 within the authorized bound. No formal within-cell tie-breaking rule was recorded.

3.6 Complete target-cell pass: 72 new entries

Software enumerated every exact Present-labelled record in Immersive × Desktop/Planar (112 memberships), Immersive × Large Display (20), and Adaptive × Large Display (12). The 144 memberships represented 131 records, including 58 prior exact identities. All entered processing; priority band, descending metadata score, and canonical identifier determined access order only. Identity reconciliation reduced them to 129 groups. Excluding already represented systems and related reports yielded 72 new systems in the cumulative selected set. The lead author specified and authorized the three target cells. The 72 new entries were the reconciliation outcome, not an inclusion quota.

4 Access and System-Level Reconciliation

September reconciliation (26 September 2026). Counts below precede the October access follow-up. Two related-report pairs and an iCAVE preprint linked to an already represented

Table S2: September selection, report access, and completed system assessments (26 September 2026). Full text, Abstract, and Unavailable count report entries; Assessed counts systems after report reconciliation. October follow-up totals are given in the text.

Set	Reports	Systems	Full text	Abstract	Unavailable	Assessed
20	20	20	20	0	0	20
18	18	18	15	0	3	15
50	50	50	25	0	25	25
200	200	197	60	1	139	57
24	24	24	1	0	23	1
72	72	72	14	0	58	14
Total	384	381	135	1	248	132

system account for the report–system difference. At the September cutoff, the 132 completed as-

sessments comprise 80 eligible, 19 unresolved, and 33 excluded/contextual systems. Unavailable reports remain access outcomes rather than scientific exclusions.

An October access follow-up supplied three previously unavailable reports within the same selected set: VIPER (Algorithmic × Desktop/Planar), NAVRNA (Immersive × Large Display, supported by tangible constrained deformation), and Cosmic View (Algorithmic × Large Display, using a planetarium dome). [66, 7, 19] These author-confirmed assessments bring the totals to 138 accessible report entries, 245 unavailable entries, one abstract-only entry, and 135 completed system assessments (83 eligible, 19 unresolved, and 33 excluded/contextual). Three additional placements bring Figure 3 to 79 represented systems and 122 placements. The six original selection sets and their membership were not rerun or expanded. The September table above remains the original dated reconciliation.

4.1 Candidate Linkage and Supported Placements

At the 26 September reconciliation, 376 selected publication entries and 373 systems had exact links to the 9,505-record candidate pool. Eight systems entered through coauthor-supplied reports or bounded additional searches for framework coverage. The 135 accessible publication entries yielded 132 completed system assessments; 249 selected systems lacked completed assessments. Related reports were grouped using the identity crosswalk, with versions, source passages, and qualifications recorded for each placement.

Among the 80 eligible systems, 76 supported 119 system–cell placements and four lacked a supported placement. Systems could occupy several cells. The 119-placement total includes Vitessce Link's subsequently confirmed measurement-based Immersive × AR/MR amendment; the original September placement ledger retained 118 supported placements. These are placement-accounting layers, not additional systems or independent validation. The lead author's later review confirmed the reported placements under the authority limits in Section 5.

4.2 Targeted Sparse-Cell Assessment

The September targeted assessment examined candidate combinations in Immersive × Desktop/Planar, Immersive × Large Display, and Adaptive × Large Display. The 144 candidate memberships resolved to 131 records and 129 system groups, representing 142 distinct system–cell cases. Thirty cases were assessed from full-text reports: 28 lacked support for the proposed placement and two remained unresolved. The other 112 cases were access-limited. At the September access cutoff, reports were unavailable for 86 of 110 Immersive–Desktop system groups, 15 of 20 Immersive–Large Display groups, and 11 of 12 Adaptive–Large Display groups. This assessment established no additional placements at that access cutoff. These case counts are distinct from the report and system denominators above.

5 Trace and Authority Limits

The report–system crosswalk at `evidence/selection-flow/unit_crosswalk.csv` records assessment set, queue identifier, route, canonical linkage, report identity and version, access, system outcome, and placement qualifications. Wave-specific scripts, frozen queues, source bindings, retrieval manifests, and assessment ledgers provide supporting trace; the retained record contains no dedicated execution logs for the static queues.

The lead author set priorities, bounds, and target cells; Codex nominated entries; deterministic scripts ranked and validated queues. We confirmed these roles against the retained session logs on 9 October 2026 (`docs/full_report_selection_authority.md`). That record, linked from `docs/reproducibility.md`, distinguishes contemporaneous task evidence from retrospective author confirmation.

Nomination authority, automated assessment, later author review, and independent validation are distinct. We did not retain documentation of separate author approval of the exact constructed queues before assessment. The lead author reviewed and confirmed the Figure 3 placements. We did not retain dated row-level approval records, and this review does not constitute independent double coding or pool-wide screening validation. The crosswalk does not provide a reason for every other candidate's non-selection.

6 Language Coverage and Evidence Limits

The initial system assessment used accessible English-language reports. Language was not a provider-query or title-and-abstract screening gate. A subsequent audit distinguished provider-reported language, title/abstract language signals, and full-report language; an English abstract was not treated as proof of an English report. The 9,505-record candidate export contains no language column.

A bounded follow-up considered seven reports from a 147-report lead set, with Trellet's French thesis considered separately. These reports have no finalized author-adjudicated eligibility and report–system reconciliation in the retained record. They therefore remain outside the approved system and placement counts. Their consideration does not establish exhaustive non-English coverage or screening reliability. Consequential translated passages require verification before any future inclusion. No provisional eligibility recommendation is presented here as an approved finding.

7 Task-Symbol Interpretation

Task letters identify navigation (N), selection/filtering (F), comparison (C), sensemaking (S), and coordination/collaborative reasoning (R). Coding follows descriptions of the implemented analytical workflow, not a demonstrated benefit or an interface affordance alone. Coordination can involve task assignment, handoff, or collective interpretation; it need not imply synchronous multiuser operation. Mechanism–environment placement evidence and task support are distinct. Retained placement rows include analytical-task descriptions, source locators, and qualifications. Repeated task letters do not establish equivalent support, performance, or evaluation across systems.

8 Per-System Evidence for the Supported Map

Tables S3 and S4 consolidate the 79 mapped systems and 122 supported assistance–modality pairings in Figure 3. They include the October access follow-up and Vitessce Link's measurement-based Immersive × AR/MR amendment. Pairings and their task letters remain separate within each system; they do not imply support for other combinations. The four eligible systems without recorded supported placements are listed separately in Table S5; they contribute to the 83 eligible systems but not the 79 mapped systems or 122 placements.

Entries summarize retained report evidence and assessment notes, not new independent validation. When retained evidence does not establish analyst controls or evaluation, the table states that assessment boundary; it does not infer that the report omits them. Placement qualifications are assessment boundaries rather than necessarily author-reported limitations. S identifiers link the compact and extended tables to `METHODS_PER_SYSTEM_EVIDENCE.csv`; its locator fields identify retained evidence passages and assessment files. R and TSC identifiers refer to the retained ledgers included with this source. Bibliographic metadata are supplied for all 79 mapped systems. The compact table uses abbreviated pairings (Alg: Algorithmic; Ad: Adaptive; Conv: Conversational; Imm: Immersive; D: Desktop/Planar; L: Large Display; V: VR; A: AR/MR; C: CAVE); task letters follow Section 7.

Evaluation types summarize retained evidence, not study quality or demonstrated benefit. Computational includes technical benchmarks and model, phantom, or output validation. Expert cases include domain-expert feedback and scientific case analyses; controlled study requires a reported comparison with human participants. Deployment requires evaluation during sustained operational use. Demonstration only denotes illustrative examples without a structured evaluation established in the retained account; not established denotes insufficient attributable evaluation evidence. Types may coexist; counts overlap and cannot be summed as a partition. Evaluation-type tally (overlapping system counts): not established 22, expert cases 29, computational 21, controlled study 5, demonstration only 6, user study 2. Deployment evaluations were not established in the retained summaries (0 classified); this is not a finding that no deployed systems exist. User study denotes a retained human study whose comparison design was insufficiently specified to classify as controlled.

Table S3: Compact overview of 79 mapped systems. Pair abbreviations are defined in Section 8; task strings follow pairing order. Evaluation types describe retained evidence and may overlap.

System	First author/year	Domain	Supported pairings	Tasks	Evaluation type
S001 5-D stem-cell correction workflow	Wait 2014	Stem-cell imaging	Alg×D	NCFS	Not established
S002 AEGIS	Zhu 2019	Functional genomics	Alg×D	NCFS	Not established
S003 Aneurysm CBR visual analytics	Spitz 2020	Aneurysm risk	Alg×D	NCFS	Expert cases
S004 AR knee-replacement visualization	Maharjan 2021	Knee surgery	Alg×A	NCFS	Computational
S005 ARGV	Drogaris 2024	3-D genomics	Alg×D; Alg×A	NCFS/NCFS	Controlled study
S006 ASCRIBE-XR	Pandolfi 2026	Scientific imaging	Alg×V; Ad×V; Conv×V	NFSR/NFSR/ NFSR	Demonstration only
S007 Augmented Reality Microscope	Jin 2022	Cancer pathology	Alg×A	CFS	Expert cases
S008 BioNet-XR	Senderin 2024	Molecular networks	Alg×D; Alg×V	NCFS/NCFS	User study
S009 BioNetApp	Roumani 2019	Gene expression	Alg×D	NCFS	Expert cases
S010 CellWhisperer	Schaefer 2025	Single-cell analysis	Alg×D; Conv×D	CFS/CFS	Computational
S011 CERA	Agraz 2024	Biomarker analysis	Alg×D; Conv×D	CFS/CFS	Not established
S012 CoCoNUT	Abouelhoda 2008	Comparative genomics	Alg×D	CFS	Computational
S013 Compressed Adjacency Matrices	Dinkla 2012	Gene regulation	Alg×D	NCFS	Expert cases
S014 Cosmic View	Duchemin 2025	Biological taxonomy	Alg×L	NCS	Not established
S015 CPA	Nguyen 2021	Pathway analysis	Alg×D	NCFS	Not established
S016 D-Rax	Nisar 2024	Chest radiology	Alg×D; Conv×D	CFS/CFS	Computational
S017 DatAR	Tanhaei 2022	Neuroscience literature	Alg×A	NCFS	Expert cases
S018 DTBIA	Yao 2025	Brain imaging	Alg×V; Imm×V	NCFS/NCFS	Expert cases
S019 Echo	Thomas 2017	Clinical workflows	Alg×L	NCFSR	Not established
S020 ExaViz	Dreher 2014	Molecular simulation	Alg×D; Ad×D; Alg×C; Ad×C; Imm×C	NCFS/NCFS/ NCFSR/ NCFSR/ NCFSR	Not established
S021 Facetto	Krueger 2020	Tissue phenotyping	Alg×D; Ad×D	NCFS/NCFS	Expert cases
S022 FathomGPT	Khanal 2024	Ocean biology	Alg×D; Conv×D	FS/FS	Computational
S023 Flud	Bharadwaj 2022	Network layout	Alg×D	FSR	Controlled study
S024 focusedMDS / distnet	Urpa 2019	Biological distances	Alg×D	CFS	Expert cases
S025 Fused DTI/HARDI	Prckovska 2011	Brain diffusion MRI	Alg×D	NCFS	Expert cases
S026 Gene Tracer	Lou 2021	Gene regulation	Alg×D	NCFS	Not established

System	First author/year	Domain	Supported pairings	Tasks	Evaluation type
S027 Generic anatomical-model builder	Xiao 2010	Anatomical modeling	Alg×D	CFS	Computational
S028 GENET	Li 2024	Molecular regulation	Alg×D	NCFS	Computational
S029 GeneVAnD	Hibbs 2005	Gene expression	Alg×D; Alg×L	NCFS/NCFSR	Expert cases
S030 i-gRINN	Serçinoğlu 2026	Protein interactions	Alg×D; Conv×D	NCFS/NCFS	Demonstration only
S031 iCAVE	Liluashvili 2017	Biological networks	Alg×D; Alg×C	NCFS/NCFS	Not established
S032 iGEAK	Choi 2019	Gene expression	Alg×D	CFS	Expert cases
S033 Interactive 3D lesion mapping	Dai 2019	Cardiac imaging	Alg×A	NCFS	Not established
S034 Interactive AI annotation in VR	Orsmaa 2025	Medical annotation	Alg×V; Ad×V; Imm×V	NCFS/NCFS/ NCFS	Controlled study
S035 LIM Tracker	Aragaki 2022	Cell tracking	Alg×D	NCFS	Computational; Expert cases
S036 Liver MR navigation	Xiang 2025	Liver surgery	Alg×A; Ad×A; Imm×A	NCFS/NCFS/ NCFS	Computational
S037 MediSyn	He 2017	Cancer evidence	Alg×D	CFS	Controlled study
S038 Metis	Menke 2024	Molecular design	Alg×D; Ad×D	FS/FS	Demonstration only
S039 MinOmics	Maes 2018	Multi-omics	Alg×L; Alg×V	NCFS/NCFS	Computational; Expert cases
S040 Molecular Dynamics Visualization (MDV)	Wiebrands 2018	Molecular dynamics	Alg×D; Alg×L	NCS/NCS	Computational; Expert cases
S041 moleculARweb Virtual Modeling Kits	Abriata 2020	Molecular simulation	Alg×D; Alg×A	NCFS/NCFS	Not established
S042 MR bone-tumor planning	Wong 2023	Bone-tumor surgery	Alg×A	NCFS	Not established
S043 MR carotid-endarterectomy planning	Colombo 2026	Vascular surgery	Alg×A	NCFS	Not established
S044 MR radiation-dose visualization	Takata 2021	Radiation exposure	Alg×A	CFS	Not established
S045 MR thalamic-surgery workflow	Zanuttini 2025	Neuro-oncology	Alg×A	NCFS	Expert cases
S046 NAVRNA	Bailly 2006	RNA structures	Imm×L	FSR	Not established
S047 NeuroBlocks	Al-Awami 2016	Connectomics	Alg×D	NFR	Expert cases
S048 Next Viewer	Günther 2026	Surgical planning	Alg×A	NCFS	User study
S049 NUI-VR2	Gao 2022	Medical segmentation	Alg×V; Imm×V	NCF/NF	Computational
S050 OncoProExp	Rahmani 2025	Cancer expression	Alg×D	CFS	Computational; Expert cases
S051 PAE Viewer	Elfmann 2023	Protein structures	Alg×D	NCFS	Expert cases
S052 Pan-cancer networks	Kling 2015	Cancer networks	Alg×D	NCFS	Not established
S053 PathwayMatrix	Dang 2015	Pathway analysis	Alg×D	NCFS	Expert cases
S054 Patient-health dashboard	Secco 2025	Patient records	Alg×D; Ad×D	NCFS/NCFS	Demonstration only
S055 PeCaX	Figaschewski 2023	Genomic variants	Alg×D	NCFSR	Computational

System	First author/year	Domain	Supported pairings	Tasks	Evaluation type
S056 PhenoFlow	Kim 2025	Stroke analysis	Alg×D; Conv×D	NCFS/NCFS	Expert cases
S057 Protein-model validation	Falk 2024	Protein validation	Alg×D	NCFS	Expert cases
S058 ProteinSketch	Ma 2026	Protein design	Alg×V; Imm×V	NFS/NFS	Computational
S059 Rapid QC-MS	Sandhu 2024	Mass spectrometry	Alg×D	CFS	Computational
S060 RATS VR planning system	Ujiie 2024	Thoracic surgery	Alg×V	NCFS	Not established
S061 REDAC	De Filippis 2026	Clinical data	Alg×D; Conv×D	CFS/CFS	Not established
S062 RetainVis	Kwon 2019	Patient prediction	Alg×D	CFS	Computational; Expert cases
S063 RNA-SeqEZPZ	Taslim 2026	Transcriptomics	Alg×D	CFS	Computational
S064 Robin	Chowdhury 2026	Genomic analysis	Alg×D; Conv×D	CFS/CFS	Not established
S065 SAMIRA	Spiegler 2025	Medical annotation	Alg×V; Ad×V; Conv×V; Imm×V	NFS/NFS/ NFS/NFS	Controlled study
S066 scSelector	Gao 2025	Single-cell analysis	Alg×D; Conv×D	CFS/CFS	Expert cases
S067 SeqPlots	Stempor 2016	Genomics	Alg×D	CFS	Computational
S068 Skin-lesion AR	Francese 2020	Dermatology	Alg×A; Imm×A	CS/CS	Computational; Expert cases
S069 Space-Time Hypercube	Fouché 2023	Embryo development	Alg×V; Imm×V	NCFS/NCFS	Expert cases
S070 SpinalSAM-R1	Liu 2025	Spine CT	Alg×D; Conv×D	CFS/CFS	Computational
S071 Surgical Theater XR platform	Steineke 2022	Neurosurgery	Alg×V; Alg×A	NCFS/NCFS	Expert cases
S072 Toxygates	Nyström-Persson 2017	Toxicogenomics	Alg×D	CFS	Expert cases
S073 VIADS	Jing 2019	Clinical hierarchies	Alg×D	NCFS	Demonstration only
S074 VIPER	St-Jean 1998	Neurosurgery	Alg×D	NCFS	Not established
S075 Virtual embolization	Sprengel 2021	Vascular intervention	Alg×D; Alg×V	NCFS/NCFS	Expert cases
S076 Virtual Island	Aurisano 2019	Animal movement	Alg×C	NCFS	Not established
S077 Vitessce Link	Mörth 2025	Spatial biology	Alg×D; Alg×A; Imm×A	NCFS/ NCFSR/CFS	Expert cases
S078 VROOM	Lau 2022	Protein structures	Alg×V	NCFS	Demonstration only
S079 VR–AR medical-image analysis workflow	Prabha 2024	Medical imaging	Alg×V; Alg×A	NCFS/NCFS	Not established

Table S4: Extended per-system evidence for the retained supported Figure 3 map. Tasks: N navigation, C comparison, F selection, S sensemaking, R collaboration. Pairings are explicit; missing evidence is labelled, not inferred.

System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S001 5-D stem-cell correction workflow [75]	Visualization and correction of automated segmentation, tracking and lineaging from 5-D stem cell image sequences. Tasks: Algorithmic × Desktop/Planar: NCFS	Automated segmentation, tracking, and lineage construction are coupled to visualization and explicit analyst correction across 5-D stem-cell sequences.	Algorithmic × Desktop/Planar	Analyst views and edits automated segmentation, tracking, and lineage results in a stereoscopic 3-D view linked to the lineage tree.	Retained abstract claims correction to 100% accuracy; validation design and independent outcomes cannot be established from that claim.	X027 Abstract 1 ¶2; X027 Conclusions ¶2
S002 AEGIS [82]	Exploratory Gene Ontology Analysis with Interactive Visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	A focus-plus-context graph layout, ontology traversal, simulations, and power studies organize GO relations and support biological investigation.	Algorithmic × Desktop/Planar	The scientist selects GO concepts and study questions; the system lays out, simulates, and computes; the scientist interprets and revises the inquiry.	Evaluation evidence was not established in the retained assessment.	bioRxiv v1 pp. 2–13; Abstract, system, and use cases Preprint is the assessed accessible version; DOI links the published report.
S003 Aneurysm CBR visual analytics [64]	Combining visual analytics and case-based reasoning for rupture risk assessment of intracranial aneurysms. Tasks: Algorithmic × Desktop/Planar: NCFS	Case-based reasoning computes similarity rankings that are linked to morphology and clinical-feature views.	Algorithmic × Desktop/Planar	Clinician selects features/weights and reviews retrieved cases; system ranks similarities; clinician assesses rupture evidence.	Six trained physicians completed evaluation tasks; limited sample and training context constrain outcome claims.	Abstract; System; Evaluation; Results

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S004 AR knee-replacement visualization [45]	A Novel Visualization System of Using Augmented Reality in Knee Replacement Surgery: Enhanced Bidirectional Maximum CorrentropyAlgorithm. Tasks: Algorithmic × AR/MR: NCFS	CT segmentation, stereo matching, WLS occlusion handling, and bidirectional maximum-correntropy ICP registration drive the report's AR overlay workflow.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Retained passage describes execution-time measurement and a comparison figure; metric definitions, baseline and outcomes unresolved.	X110 PDF pp. 9-14, Proposed System, Proposed Equation, Results and Discussion Implemented on sampled CT images and online surgical videos; this does not establish intraoperative clinical benefit.
S005 ARGV [18]	ARGV: 3D genome structure exploration using augmented reality. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × AR/MR: NCFS	The web system transforms chromosome-conformation data into interactive 3D genome structures exposed on both planar and AR displays.	Algorithmic × Desktop/Planar; Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	A comparative user study is mentioned in the retained abstract; participants, design and numerical results not established in the retained assessment.	X021 Results > ARGV is a 3D genome study environment > Manipulating 3D genomes and annotations ¶1; X021 Results > ARGV is a 3D genome study environment > Exploring a 3D chromosome structure in augmented reality ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S006 ASCRIBE-XR [52]	Turning Immersive Viewers into Analytical Workspaces: ASCRIBE-XR and Agent-Driven Scientific Visualization. Tasks: Algorithmic × VR: NFSR; Adaptive × VR: NFSR; Conversational × VR: NFSR	The implemented agent invokes segmentation, meshing, resolution, and rendering tools inside the immersive workflow. | The scientific request and loaded data are triggers; the agent resolves unstated method/resolution choices and changes its tool sequence. | Open-ended language specifies a scientific intent that the agent operationalizes across multiple analysis and rendering tools.	Algorithmic × VR; Adaptive × VR; Conversational × VR	Scientist states a goal; the agent resolves operational choices and executes a workflow; scientist inspects and can refine the output.	Implemented architectural demonstrations, including a Panicum hallii root use case; no participant study or scientific-outcome comparison.	Abstract; Fig. 1; §2 ¶2–4; Appendix A Table A2; §2 ¶2–4; Appendix A Table A2; Discussion ¶2–3; Abstract; Fig. 1; Appendix A Table A2 Implemented examples exceed parameter editing, although benefit was not evaluated. | The mechanism resolves analytic operations rather than mapping speech to a fixed command.
S007 Augmented Reality Microscope [30]	Independent assessment of a deep learning system for lymph node metastasis detection on the Augmented Reality Microscope. Tasks: Algorithmic × AR/MR: CFS	A deep-learning metastasis detector supplies case-level overlays through an augmented-reality microscope while pathologists inspect lymph-node tissue.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Pathologist re-review of selected false-negative fields and annotation errors; no analytical-benefit conclusion inferred.	X014 PDF p. 2
S008 BioNet-XR [60]	BioNet-XR: Biological Network Visualization Framework for Virtual Reality and Mixed Reality Environments. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × VR: NCFS	Desktop BioNet-XR computes layouts, shortest paths, and PageRank/Steiner/ random-walk subnetworks. | The VR version implements analysis functions similar to the desktop build for network exploration.	Algorithmic × Desktop/Planar; Algorithmic × VR	Analyst chooses nodes/algorithms; software computes layouts, paths, and subnetworks; analyst interprets network structure.	Desktop-only usability study with 30 participants; VR/MR benefit was not tested because participants saw videos only.	PDF pp. 12–16, 22; Conclusion

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S009 BioNetApp [56]	BioNetApp: An interactive visual data analysis platform for molecular expressions. Tasks: Algorithmic × Desktop/Planar: NCFS	Correlation, comparison, clustering, outlier detection, and curve fitting are computed within coordinated molecular-expression views.	Algorithmic × Desktop/Planar	Analyst selects experiments and filters; algorithms mine correlations/clusters; analyst inspects time-course patterns.	Alcohol-induced fatty-liver case study; no participant or baseline study.	Abstract; Results and discussion; Conclusions ¶1–4
S010 CellWhisperer [58]	Multimodal learning enables chat-based exploration of single-cell data. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	A multimodal language model maps dialogue about single-cell data to analytical operations and visual results.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Retained passage introduces biological-characteristic prediction tests; design and outcomes not established in the retained assessment. Chat availability is not proof of analytic benefit.	X030 PDF p. 1; X030 PDF p. 1
S011 CERA [3]	ChatGPT-Enhanced ROC Analysis (CERA): A shiny web tool for finding optimal cutoff points in biomarker analysis. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	The Shiny application computes ROC cutoffs and uses an LLM to mediate explanations and analysis choices.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X040 Discussion ¶1
S012 CoCoNUT [1]	CoCoNUT: an efficient system for the comparison and analysis of genomes. Tasks: Algorithmic × Desktop/Planar: CFS	Rare matches, chaining, alignment, and comparative plots reveal genomic similarity, rearrangement, and duplication.	Algorithmic × Desktop/Planar	Analyst sets the comparative problem and parameters; algorithms generate chains/alignments; analyst inspects genomic structure.	Benchmarks and comparisons with BioMart and prior synteny results; no user study.	Results > Mammalian chromosomes ¶1–8; A. thaliana ¶1; Conclusion

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S013 Compressed Adjacency Matrices [16]	E. coli and B. subtilis regulatory networks; Inspect motifs, paths, and neighborhoods in gene-regulatory networks. Tasks: Algorithmic × Desktop/Planar: NCFS	Structure-aware cutting, ordering, and compression reorganize a gene-regulatory-network matrix before interactive highlighting, filtering, and path or neighborhood inspection.	Algorithmic × Desktop/Planar	Inspect motifs, paths, and neighborhoods in gene-regulatory networks. Interaction consequence: Reorganizes and navigates the representation	Network demonstrations and informal biologist feedback Limitations: No controlled insight study; report doubts routine use by biologists; benefit boundary: Qualitative structural visibility	Local thesis PDF pp. 41-42, 55-61; article Sections 3-6 The algorithm structures the view; interaction does not revise the organizing algorithm.
S014 Cosmic View [19]	Hierarchical biological relationships and similarities. Tasks: Algorithmic × Large Display: NCS	Constrained multidimensional scaling and spherical decluttering organize biological similarities for inspection.	Algorithmic × Large Display	Scientist explores organized biological relationships. Detailed scoping, editing and undo controls not established in the retained assessment.	Evaluation evidence was not established in the retained assessment.	DUCHEMIN_2025_cosmic_view_life_on_earth.pdf, pp. 93–96, 99–101 Large Display pairing is the planetarium dome; no extra Desktop, CAVE or Immersive pairing inferred.
S015 CPA [48]	CPA: a web-based platform for consensus pathway analysis and interactive visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	Eight pathway methods, cross-dataset meta-analysis, and consensus computation feed interactive pathway and gene-relation views.	Algorithmic × Desktop/Planar	The analyst chooses methods and datasets; CPA computes and combines results; the analyst compares pathways and interprets biological meaning.	Evaluation evidence was not established in the retained assessment.	PMC8262702 XML; Abstract, Introduction, Results, and graphical abstract Established methods are integrated substantively; no Adaptive mechanism is inferred.
S016 D-Rax [49]	D-Rax: Domain-specific Radiologic assistant leveraging multi-modal data and eXpert model predictions. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	The radiologic assistant combines images with expert-model predictions and mediates diagnostic exploration through language.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Retained passage includes a truncated numerical table and discussion of expert-model-guided training; exact metrics/comparisons cannot be reliably recovered here.	X035 PDF p. 2; X035 PDF p. 10

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S017 DatAR [70]	Exploring Neuroscience Literature and Understanding Relations Between Brain-Related Topics - Using Augmented Reality. Tasks: Algorithmic × AR/MR: NCFS	Literature-derived brain-topic relations are organized as a graph and explored through an AR brain representation.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Retained passage describes neuroscientist interviews and iterative relation-finding design; exact sample and outcome measures not established in the retained assessment.	X189 PDF p. 2; X189 PDF p. 2
S018 DTBIA [80]	DTBIA: An Immersive Visual Analytics System for Brain-Inspired Research. Tasks: Algorithmic × VR: NCFS; Immersive × VR: NCFS	3D force-directed edge bundling, top-connection selection, threshold filtering, animation, and hierarchical region-to-voxel-to-slice analysis organize functional and structural brain data. | The system couples real-scale and large-scale brain models with walking, flying, teleportation, and controller-based spatial navigation while computational filtering and bundling guide multiscale inspection.	Algorithmic × VR; Immersive × VR	Specific analyst controls were not established in the retained assessment.	Expert cases are mentioned in retained manuscript prose; sample, design and findings not established in the retained assessment. No controlled estimate of an immersive advantage.	Published PDF pp. 5-8; Sections IV.A, IV.C, and V; Published PDF pp. 4-8; Requirement R1 and Section IV.B Exploration in an Immersive Environment The paper evaluates two expert case studies, not a controlled comparison of scientific outcomes. | Immersive placement rests on multiscale spatial navigation coupled to filtering and bundling, not on the headset alone.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S019 Echo [72]	Echo: A Large Display Interactive Visualization of ICU Data for Effective Care HandOffs. Tasks: Algorithmic × Large Display: NCFSR	Patient summaries, normalcy scores, and threshold-based automatic anomaly detection are presented on a tiled wall for team comparison, annotation, and handoff discussion.	Algorithmic × Large Display	Clinicians share inspection of computed patient summaries and anomaly highlighting during ICU handoff; detailed computational scoping/edit/undo controls not established in the retained assessment.	Evaluation evidence was not established in the retained assessment.	PDF pp. 1-2 and 5-7; Section 3, Anomaly Detection; Figs. 1, 4, and 7 Algorithmic assistance is simple threshold comparison, not Adaptation; evaluation is small and does not establish clinical outcome benefit.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S020 ExaViz [17]	ExaViz: a flexible framework to analyse, steer and interact with molecular dynamics simulations. Tasks: Algorithmic × Desktop/Planar: NCFS; Adaptive × Desktop/Planar: NCFS; Algorithmic × CAVE: NCFSR; Adaptive × CAVE: NCFSR; Immersive × CAVE: NCFSR	Desktop ExaViz combines molecular-dynamics analysis, content-aware camera paths, semi-automatic viewpoints, and interactive steering. | User-selected molecular targets and geometry drive least-occluded viewpoints and task-aware paths; recorded steering targets can drive a later automatic run. | The same molecular computations, guided navigation, analysis graphs, and steering operate in the implemented CAVE-like environment. | Target selection and molecular geometry tailor viewpoints and paths; a captured human steering path controls later automatic steering with manual correction. | Two head-tracked views, co-located pointing, spatial atom picking, and a haptic device that follows the user couple assistance to embodied CAVE state.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar; Algorithmic × CAVE; Adaptive × CAVE; Immersive × CAVE	The scientist selects targets and initiates steering; the system proposes views and executes/replays forces; the scientist observes, overrides, corrects, and interprets.	Evaluation evidence was not established in the retained assessment.	HAL report pp. 8–15; §§3.6–5.1; HAL report pp. 9–15; §§4.1–5.1; HAL report pp. 8–15; §§3.6–5.1 and Fig. 7; HAL report pp. 10–15; §§4.1, 4.3, and 5.1; HAL report pp. 12–15; §§4.2–5.1 and Fig. 7 The framework demonstrates technical capability; controlled scientific benefit is not established. | Data/task-conditioned tailoring and interaction-history replay qualify; ordinary parameters alone are not the basis. | One system occupies multiple modalities; this is not a second system count. | Implemented feedback/history loop; “adapt” wording about human behavior is not itself the evidence. | Immersive placement rests on tracked spatial assistance and haptics, not CAVE display alone.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S021 Facetto [35]	Multiplex tissue images and cell features; Discover and correct cell phenotypes. Tasks: Algorithmic × Desktop/Planar: NCFS; Adaptive × Desktop/Planar: NCFS	EM clustering, UMAP, and CNN classification are integrated with coordinated image, projection, distribution, table, and phenotype-tree views. | Experts select and label phenotype subsets; clustering results train a classifier whose later assignments change, creating a user-feedback-to-model loop.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar	Discover and correct cell phenotypes. Interaction consequence: Changes clusters/labels and feeds revised labels into later classification	Ten interviews and two guided expert case studies Limitations: Small samples; authors operated UI; qualitative outcomes; tree lacks fuzzy membership; benefit boundary: Experts could direct phenotype analysis in cases	Related preprint PDF pp. 1, 3-5; Fig. 1 and Fig. 3; Related preprint PDF pp. 2-5; Interactive Clustering and Classification Published identity is linked to an accessible related preprint; active learning is future work. | Adaptive placement rests on demonstrated classifier retraining, not on parameter adjustment; experts directed cases while authors operated the interface.
S022 FathomGPT [32]	FathomNet observations and images; Explore ocean-science data through natural-language intentions. Tasks: Algorithmic × Desktop/Planar: FS; Conversational × Desktop/Planar: FS	The system resolves scientific names, constructs and executes SQL or Python, retrieves results, and generates interactive charts from user requests. | Natural-language subgoals and conversation history mediate query construction, result retrieval, and chart refinement in the planar interface.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Explore ocean-science data through natural-language intentions. Interaction consequence: Changes queries, displays, and returned data	2,977 workshop prompts; 781-prompt success analysis; context ablation Limitations: Not a controlled scientific-decision or adoption study; phrasing and data limitations; benefit boundary: 72.08% prompt success; context improved correctness in reported ablation	Author report PDF pp. 2-3; System architecture and Fig. 2; Author report PDF pp. 2-3 and 9-14; Fig. 1 and System architecture Prompt-level success is not evidence of improved scientific decisions. | Conversation supports bounded subgoals; it does not establish durable goal learning or autonomous scientific agency.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S023 Flud [8]	Biological interaction networks; Produce biologically useful network layouts. Tasks: Algorithmic × Desktop/Planar: FSR	A multi-criterion layout score, algorithmic clues, and simulated annealing alternate with crowd node placements to improve biological-network layouts.	Algorithmic × Desktop/Planar	Produce biologically useful network layouts. Interaction consequence: Human moves and algorithm search alter the layout; no preference learning	Roughly 2,000 crowd workers and seven biologically trained participants in published study Limitations: Layout/task metrics do not demonstrate biological discovery; benefit boundary: Higher layout scores and task effectiveness	Accessible report PDF pp. 1-4 and 17-21; Sections 1 and 2.1 The system does not learn an individual user's preferences; layout improvements do not establish better biological discovery.
S024 focusedMDS / distnet [74]	Focused multidimensional scaling: interactive visualization for exploration of high-dimensional data. Tasks: Algorithmic × Desktop/Planar: CFS	FocusedMDS and distnet recompute local projections and distance relations around analyst-selected objects.	Algorithmic × Desktop/Planar	Analyst focuses objects and compares views; algorithms recompute local projections/networks; analyst judges distortion and structure.	Worked biological examples; no participant study.	Abstract; Results > distnet ¶1–3; focusedMDS ¶1–4
S025 Fused DTI/HARDI [54]	Fused DTI/HARDI Visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	A classification scheme automatically selects DTI or HARDI per voxel; GPU glyphs and tractography expose the fused result interactively.	Algorithmic × Desktop/Planar	The analyst chooses views and encodings; the system selects voxel models and renders; the analyst interprets anatomy and can adjust display controls.	Qualitative user evaluation retained; design/sample/outcomes not established in the retained assessment. Scientific-outcome improvement not established.	Published PDF pp. 1–2, 4–10; Abstract, §2.3, framework, and evaluation Qualitative user evaluation supports utility, not scientific-outcome improvement.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S026 Gene Tracer [42]	Gene Tracer: a smart, interactive, voice-controlled Alexa skill For gene information retrieval and browsing, mutation annotation and network visualization. Tasks: Algorithmic × Desktop/Planar: NCFS	Gene retrieval, mutation annotation, and network construction are integrated with an interactive planar visualization; the voice layer remains a boundary case.	Algorithmic × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X044 3 Results ¶1; X044 1 Introduction ¶3
S027 Generic anatomical-model builder [79]	Building generic anatomical models using virtual model cutting and iterative registration. Tasks: Algorithmic × Desktop/Planar: CFS	Interactive model cutting defines subvolumes that rigid, affine, and deformable registration algorithms align and average.	Algorithmic × Desktop/Planar	Analyst specifies the subvolume; software executes registration and averaging; analyst interprets group-shape differences.	No participant study; registration sharpness, Dice similarity, and RMSE are reported.	Abstract; Methods > Overview ¶1; Results > Validation ¶1–5
S028 GENET [38]	Development and Validation of an AI-Driven System for Automatic Literature Analysis and Molecular Regulatory Network Construction. Tasks: Algorithmic × Desktop/Planar: NCFS	NLP named-entity and relation models construct a filterable molecular-regulatory knowledge graph for mechanism search and hypothesis generation.	Algorithmic × Desktop/Planar	The system extracts relations and constructs the graph; the researcher searches, filters, interprets mechanisms, and validates selected hypotheses experimentally.	Computational comparison of entity/relation extraction against existing methods, with sequencing and immunofluorescence validation of a molecular prediction; no controlled analyst-benefit study established. Supplement evaluation check: published report Abstract and Sections 2.1 and 2.3.	PMC11600262 XML; Abstract, Development Framework, network construction, and validation Model accuracy and one checked pathway support capability, not broad decision benefit.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S029 GeneVAnD [29]	Visualization methods for statistical analysis of microarray clusters. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × Large Display: NCFSR	Statistical cluster summaries and linked gene views are computed and coordinated on conventional screens. | The same linked analytical views are shown and used together on a display wall.	Algorithmic × Desktop/Planar; Algorithmic × Large Display	Analyst chooses clusters and linked views; statistical routines compute summaries; analyst interprets gene patterns.	Worked microarray examples; no controlled user study.	Results > Multiple simultaneous views ¶2; Fig. 9
S030 i-gRINN [61]	i-gRINN: a next-generation web server for protein energy network analysis of heterogeneous biomolecular systems with natural language-based data exploration. Tasks: Algorithmic × Desktop/Planar: NCFS; Conversational × Desktop/Planar: NCFS	Protein-energy networks are computed and queried through a web server whose natural-language layer mediates data exploration.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Acknowledgements mention beta testing and feedback; no formal evaluation design or outcomes extracted.	X037 Abstract 1 ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S031 iCAVE [40, 39]	iCAVE: an open source tool for visualizing biomolecular networks in 3D, stereoscopic 3D and immersive 3D | iCAVE: an open source tool for immersive 3D visualization of complex biomolecular interaction networks. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × CAVE: NCFS	The same 3D layout, clustering, edge-bundling, network-query, and metric functions are analytically presented in a desktop screen environment. | 3D layout, clustering, edge bundling, network metrics, and database-generated networks are presented and explored in a projection-based CAVE environment.	Algorithmic × Desktop/Planar; Algorithmic × CAVE	Specific analyst controls were not established in the retained assessment.	Evaluation evidence was not established in the retained assessment.	Published PDF pp. 1, 3, and 5-11; Abstract and System description; Published PDF pp. 1, 3, and 5-11; Abstract, Fig. 1d, and Network layout sections Desktop/Planar is the frozen modality for rotatable 3D shown on a planar screen; stereoscopic desktop glasses do not create a sixth modality. | Established as Algorithmic × CAVE. Immersive assistance is not inferred because the report does not show the assistance algorithm changing with tracked CAVE context.
S032 iGEAK [9]	iGEAK: an interactive gene expression analysis kit for seamless workflow using the R/shiny platform. Tasks: Algorithmic × Desktop/Planar: CFS	An interactive Shiny pipeline computes normalization, differential expression, clustering, and enrichment views.	Algorithmic × Desktop/Planar	Analyst explicitly chooses methods and parameters; the pipeline recomputes results; analyst compares and interprets them.	Worked datasets and workflow demonstration; no controlled user study.	Abstract; Results > Tabs ¶1, ¶4–15

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S033 Interactive 3D lesion mapping [13]	Neurosurgical Craniotomy Localization Using Interactive 3D Lesion Mapping for Image-Guided Neurosurgery. Tasks: Algorithmic × AR/MR: NCFS	Image registration and 3D lesion mapping transform patient scans into an interactive AR localization view.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X108 PDF p. 2; X108 PDF p. 2
S034 Interactive AI annotation in VR [51]	Interactive AI annotation of medical images in a virtual reality environment. Tasks: Algorithmic × VR: NCFS; Adaptive × VR: NCFS; Immersive × VR: NCFS	A deep model converts sparse corrective clicks into a revised whole-volume segmentation in VR. | Radiologist corrective clicks are the trigger; the learned segmentation changes across the volume after each feedback cycle. | 3D click locations in the immersive volume become positive/negative model prompts that alter segmentation.	Algorithmic × VR; Adaptive × VR; Immersive × VR	Radiologist supplies corrective examples; the model revises a volume-wide segmentation; radiologist reviews and corrects the result.	Three radiologists, 50 scans, 192 annotation tasks; quality improved for two radiologists but not the third; unlocalized changes remained a concern.	Interactive annotation ¶1, ¶3; Apparatus > VR software ¶2–3; Interactive annotation ¶1, ¶3; Results; Apparatus > VR software ¶2–3; Interactive annotation ¶1, ¶3 This feedback-to-model loop exceeds direct local editing. | The spatial input changes computation, not just camera pose.
S035 LIM Tracker [5]	LIM Tracker: a software package for cell tracking and analysis with advanced interactivity. Tasks: Algorithmic × Desktop/Planar: NCFS	Automatic tracking results are visualized, corrected, and rerun in a linked microscopy workflow.	Algorithmic × Desktop/Planar	Analyst inspects and corrects tracks; the system reruns tracking; analyst validates lineage continuity.	Worked cases and SEG/DET/TRA benchmark metrics; no controlled analyst study.	Case 3 ¶3–6; Evaluation

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S036 Liver MR navigation [78]	Comprehensive mixed reality surgical navigation system for liver surgery. Tasks: Algorithmic × AR/MR: NCFS; Adaptive × AR/MR: NCFS; Immersive × AR/MR: NCFS	Rigid registration and LIBR deformation correction transform intraoperative surface evidence into an updated liver model. | New tracked intraoperative surface data are the trigger; the registration intervention changes the preoperative model displayed in HoloLens. | Tracked stylus/headset coordinates and the local surgical surface resolve spatial correspondence and update the overlaid model.	Algorithmic × AR/MR; Adaptive × AR/MR; Immersive × AR/MR	Surgeon supplies a surface or tracked points; registration updates the model; surgeon interprets the corrected spatial relation.	Phantom validation across five deformation states; no clinical outcome study.	Abstract; Method > Comprehensive system ¶1; Fig. 1; Results; Introduction ¶5–6; Method > Comprehensive system ¶1; Fig. 1; Introduction ¶2, ¶5–6; Method > Comprehensive system ¶1 This is data/context-conditioned model correction, not direct parameter adjustment. | Spatial context determines the analytic registration/correction, beyond maintaining a view.
S037 MediSyn [28]	Linked CGI and DTC biomedical datasets; Compare drugs, targets, tumor types, provenance, and conflicting evidence. Tasks: Algorithmic × Desktop/Planar: CFS	Linked biomedical datasets are algorithmically synthesized into matrix layers; sorting, uncertainty encodings, and provenance links support comparison and evidence inspection.	Algorithmic × Desktop/Planar	Compare drugs, targets, tumor types, provenance, and conflicting evidence. Interaction consequence: Changes evidence presentation and ordering; not model state	Six-person counterbalanced drug-selection and inconsistency tasks Limitations: Small sample, simplified tasks, limited domain familiarity; no clinical-decision endpoint; benefit boundary: Faster task completion with correct answers in study	Background; Overview of MediSyn; Figs. 1 and 5-7 The six-person study supports simplified evidence-selection tasks, not improved treatment decisions.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S038 Metis [46]	Generated molecules and medicinal-chemistry feedback; Teach molecular generators detailed expert preferences. Tasks: Algorithmic × Desktop/Planar: FS; Adaptive × Desktop/Planar: FS	The GUI shows generated molecules, similar active compounds, and model explanations while converting structured feedback into reward-model or reward-function inputs. | Chemist feedback trains a reward model or updates a constructed reward function; a REINVENT loop reloads newly generated molecules for further feedback.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar	Teach molecular generators detailed expert preferences. Interaction consequence: Changes feedback data, reward model, and later generated outputs	Technical implementation and fictitious MAPK10 examples Limitations: No participant or outcome experiment; desktop and REINVENT coupling; benefit boundary: No evaluated benefit reported	Application overview; Molecular display; Feedback to de novo design; Feedback to de novo design; Figs. 1-3 The report demonstrates the mechanism with fictitious examples and no participant or outcome study. | Adaptive placement rests on feedback changing model scoring and later outputs, not on simple parameter editing; alignment benefit is proposed rather than evaluated.
S039 MinOmics [44]	MinOmics, an Integrative and Immersive Tool for Multi-Omics Analysis. Tasks: Algorithmic × Large Display: NCFS; Algorithmic × VR: NCFS	Database queries, network analysis, and structural comparison are jointly displayed on an 8.3 m^2 wall. | The redox-proteomics use case superposes selected protein structures in a VR headset for fold comparison.	Algorithmic × Large Display; Algorithmic × VR	Analyst selects datasets and structural subsets; queries and models are computed; analyst compares folds and formulates biological interpretations.	Performance benchmarks and a scientific use case; formal user studies explicitly not yet conducted.	Results > Scientific Applications > Use Case 3 ¶2

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S040 Molecular Dynamics Visualization (MDV) [76]	Molecular Dynamics Visualization (MDV): Stereoscopic 3D Display of Biomolecular Structure and Interactions Using the Unity Game Engine. Tasks: Algorithmic × Desktop/Planar: NCS; Algorithmic × Large Display: NCS	MDV loads trajectories and recomputes secondary-structure representations in real time on its desktop build. | The same trajectory and secondary-structure computations drive the HIVE cylindrical stereoscopic display.	Algorithmic × Desktop/Planar; Algorithmic × Large Display	Analyst loads and manipulates molecular trajectories; Stride recomputes secondary structure; analyst interprets changing conformations.	Technical performance and application cases; no comparative analyst-outcome study.	Implementation ¶1, ¶3; User Interface ¶1; Application ¶2; Discussion ¶1–3
S041 moleculARweb Virtual Modeling Kits [2, 12]	Building blocks for commodity augmented reality-based molecular visualization and modeling in web browsers. | Visualization, Interactive Handling and Simulation of Molecules in Commodity Augmented Reality in Web Browsers Using moleculARweb's Virtual Modeling Kits. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × AR/MR: NCFS	Implemented browser tools compute molecular mechanics and experimental observables while users manipulate or navigate molecular models in the documented planar or AR workflow.	Algorithmic × Desktop/Planar; Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X012 AR-based modeling of biological macromolecules paragraphs 3-5; X128 PDF pp. 2 and 5 Implemented prototype research tasks are supported; broader research utility and benefit remain unevaluated.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S042 MR bone-tumor planning [77]	Mixed Reality Improves 3D Visualization and Spatial Awareness of Bone Tumors for Surgical Planning in Orthopaedic Oncology: A Proof of Concept Study. Tasks: Algorithmic × AR/MR: NCFS	Patient-specific bone-tumor models are reconstructed and manipulated in MR to compare spatial relationships and surgical plans.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X184 Abstract 1 ¶1
S043 MR carotid-endarterectomy planning [11]	Mixed reality applied to surgical planning and tailoring of carotid endarterectomies. Tasks: Algorithmic × AR/MR: NCFS	Patient-specific vascular models are segmented and manipulated in MR to compare anatomy and tailor the operative plan.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X094 Discussion ¶3
S044 MR radiation-dose visualization [69]	Mixed Reality Visualization of Radiation Dose for Health Professionals and Patients in Interventional Radiology. Tasks: Algorithmic × AR/MR: CFS	Radiation-dose calculations are registered to the interventional scene so clinicians can inspect and compare exposure patterns in MR.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X088 Abstract 1 ¶1
S045 MR thalamic-surgery workflow [81]	Role of mixed reality in the workflow of complex neurooncological surgeries: a case analysis in thalamic surgery. Tasks: Algorithmic × AR/MR: NCFS	Multimodal imaging and segmented tumor/anatomy models are combined in MR for thalamic-surgery planning and workflow review.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Retained abstract describes a retrospective 2022–2024 surgical cohort with clinical/radiological data; sample and outcomes not established in the retained assessment.	X095 Abstract 1 ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S046 NAVRNA [7]	From Visualization to Manipulation of RNA Secondary and Tertiary Structures. Tasks: Immersive × Large Display: FSR	Tracked tabletop tokens specify moved nodes and anchors; constrained deformation preserves edge lengths and handles collisions.	Immersive × Large Display	Scientists position tracked tabletop tokens to choose moved nodes and anchors; the system computes constrained deformation.	Evaluation evidence was not established in the retained assessment.	BAILLY_2006_ RNA_structures_ NAVRNA.pdf, pp. author PDF pp. 2, 5–6; sections 3 and 5.2 Immersive pairing limited to embodied constraints on deformation, not pan/zoom or display size. October target_ group_id differs from persisted September group; exact canonical ID/DOI binds this report.
S047 NeuroBlocks [4]	Large electron-microscopy volumes and segmentation states; Coordinate segmentation and proofreading across large connectomics projects. Tasks: Algorithmic × Desktop/Planar: NFR	Automatic RhoANA segmentation is integrated with manual and semi-automatic proofreading, task assignment, comparison, provenance, approval, and restoration views.	Algorithmic × Desktop/Planar	Coordinate segmentation and proofreading across large connectomics projects. Interaction consequence: Changes segments, task assignments, approvals, and restored workflow states	Two neuroscientist collaborations including a 955-GB dataset and proofreading case Limitations: Qualitative evidence; no controlled productivity or scientific-outcome comparison; benefit boundary: Qualitative utility in cases	Author report PDF pp. 1 and 5-9 Qualitative case studies do not quantify productivity or scientific-outcome gains.
S048 Next Viewer [27]	Next viewer–development of a software for the spatial computing data glasses apple vision pro for patient-specific preoperative planning and navigation. Tasks: Algorithmic × AR/MR: NCFS	Patient-specific imaging is segmented and spatially rendered in Apple Vision Pro for preoperative inspection and planning.	Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Retained evaluation passage reports overall score +1.7 and participant feedback. Scale, design and sample not established in the retained assessment; latency, learning curve and workflow integration limits are mentioned.	X026 Methods > Study design and evaluation of the app ¶1; X026 Introduction > Objective ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S049 NUI-VR2 [26]	Vessel and tumor volumes; Delineate and inspect medical volumes. Tasks: Algorithmic × VR: NCF; Immersive × VR: NF	User-placed 3D seeds feed SVM-based feature selection and segmentation that creates a probability volume for assisted volume rendering. | The segmentation pipeline consumes spatial seeds placed with in-environment gestures and returns a probability volume that users inspect in the same immersive 3D workspace.	Algorithmic × VR; Immersive × VR	Delineate and inspect medical volumes. Interaction consequence: Changes segmentation parameters and rendered outputs; not adaptive learning	Mouse-versus-NUI initialization on vessel and four tumors using Dice/Hausdorff Limitations: No participant study or separate usability/decision-benefit measure; benefit boundary: Technical segmentation comparison	Published PDF pp. 1-6; Introduction and System design; Published PDF pp. 1-6; Introduction and NUI interaction design Speech commands are an input channel, not evidence of Conversational or Adaptive assistance. | Immersive placement rests on spatial seed specification coupled to segmentation, not on VR display alone; no participant study establishes usability or clinical decision benefit.
S050 OncoProExp [55]	OncoProExp: An interactive shiny web application for comprehensive cancer proteomics and phosphoproteomics analysis. Tasks: Algorithmic × Desktop/Planar: CFS	Statistical and pathway analyses are integrated with interactive cancer proteomics views.	Algorithmic × Desktop/Planar	Analyst selects cohorts and analyses; software computes differential/pathway results; analyst interprets molecular patterns.	Worked cancer datasets and technical validation; no controlled analyst-benefit study.	Abstract; Methods; Results; Discussion

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S051 PAE Viewer [20]	PAE viewer: a webserver for the interactive visualization of the predicted aligned error for multimer structure predictions and crosslinks. Tasks: Algorithmic × Desktop/Planar: NCFS	Linked predicted-error, sequence, structure, and cross-link views support inspection of protein-model reliability.	Algorithmic × Desktop/Planar	Analyst selects residues/regions and cross-links; system coordinates uncertainty and structure views; analyst judges model reliability.	Worked protein examples; no controlled user study.	Abstract; Implementation > Overview and representation; Examples; Conclusion
S052 Pan-cancer networks [34]	Efficient exploration of pan-cancer networks by generalized covariance selection and interactive web content. Tasks: Algorithmic × Desktop/Planar: NCFS	Generalized sparse inverse covariance selection integrates cancers and data types, then exposes inferred networks as linked interactive web content.	Algorithmic × Desktop/Planar	The analyst selects cancers and network regions; the system infers and exposes links; the analyst interprets modules against pathway and drug evidence.	Evaluation evidence was not established in the retained assessment.	PMC4551906 XML; Abstract, Introduction, Methods, and analysis examples Method robustness and rediscovered mechanisms do not establish user-benefit causality.
S053 PathwayMatrix [14]	PathwayMatrix: visualizing binary relationships between proteins in biological pathways. Tasks: Algorithmic × Desktop/Planar: NCFS	Ordering, grouping, hierarchy, and lenses reorganize pathway relationships for comparative inspection.	Algorithmic × Desktop/Planar	Analyst selects pathways and lenses; software orders and groups relations; analyst compares patterns and traces candidates.	Feedback from three domain experts; a new cross-pathway relation was found, while binary decomposition limits causal/temporal reasoning.	Methods > Ordering, lensing, grouping, hierarchy; Expert feedback ¶1–5

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S054 Patient-health dashboard [59]	A modular visual analytics dashboard for patient health data. Tasks: Algorithmic × Desktop/Planar: NCFS; Adaptive × Desktop/Planar: NCFS	Transformer LLMs extract and classify clinical entities, which feed interactive medication, laboratory, and highlighted-text modules. | The dashboard dynamically selects modules from document content type and clinical relevance, and the clinician can correct extraction errors.	Algorithmic × Desktop/Planar; Adaptive × Desktop/Planar	The clinician initiates record review and corrects extraction; the system extracts, categorizes, and selects modules; the clinician interprets the resulting history.	Synthetic-record demonstrations; no structured clinical evaluation retained. Clinical benefit not established.	Published PDF pp. 1–2, 6–13; Abstract, architecture, and modules; Published PDF pp. 1–2, 9–13; Abstract and interface workflow Implemented capability; synthetic records and no structured clinical evaluation limit benefit claims. | Data-conditioned tailoring qualifies; future user-context personalization is not claimed.
S055 PeCaX [22]	Somatic variants, clinical annotations, pathways, drugs, and literature; Annotate variants and develop treatment hypotheses. Tasks: Algorithmic × Desktop/Planar: NCFSR	ClinVAP variant annotation, pathway and drug-network extraction, and automatic layout feed interactive tables and networks used in molecular-tumor-board inspection.	Algorithmic × Desktop/Planar	Annotate variants and develop treatment hypotheses. Interaction consequence: Changes evidence selection, annotation, and presentation	Processing-time benchmark Limitations: No quantitative usability or clinical-decision study; usability study is future work; benefit boundary: Processing performance only	Introduction; Methods - Clinical annotation, Network generation, GUI Processing performance is reported; a usability study is future work.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S056 PhenoFlow [33]	Clinical cohort exploration. Tasks: Algorithmic × Desktop/Planar: NCFS; Conversational × Desktop/Planar: NCFS	An LLM synthesizes executable cohort-construction code from metadata; temporal folding and linked cohort and patient views expose results for supervision and exploration. | Neurologists define and refine cohorts in natural language; the LLM normalizes terminology, synthesizes executable operations, and returns explanations and small-multiple diagnostics for supervision.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Neurologist specifies a natural-language cohort request and can inspect generated wrangling code, explanations and field views. Scientific interpretation remains with the analyst; undo controls not established in the retained assessment.	Design collaboration with two neurologists and clinical case studies during iterative refinement; no controlled estimate of calibration improvement. Explanations could conceal errors or omit needed information.	Abstract; Section 5, especially 5.1-5.2; report pp. 9-12; Section 5.1.1 Natural Language Cohort Construction; report pp. 9-13 Not placed as Adaptive: changing generated code in response to a direct request is not by itself system-selected adaptation. | Natural language mediates analytic operations and explanation; the authors describe the LLM as fragile and keep neurologists in the supervisory loop.
S057 Protein-model validation [21]	A Visual Environment for Data Driven Protein Modeling and Validation. Tasks: Algorithmic × Desktop/Planar: NCFS	Validation metrics and local quality measures are computed and coordinated across heatmap, molecular, and statistical views.	Algorithmic × Desktop/Planar	The expert chooses the model and validation questions; metrics guide attention; the expert inspects linked evidence and decides what to repair.	Retained placement qualification explicitly reports no controlled user-benefit study; examples demonstrate capability only.	Author manuscript pp. 1, 4–10; Abstract, framework, and examples Examples demonstrate capability; no controlled user-benefit study is reported.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S058 ProteinSketch [43]	Protein backbones, volumes, sequences, structures, and binders; Express three-dimensional topology and envelope constraints for protein design. Tasks: Algorithmic × VR: NFS; Immersive × VR: NFS	RFdiffusion, ProteinMPNN, structure prediction, and filtering refine user-specified protein topologies and volumetric envelopes. | Bare-hand bimanual sketching encodes position, orientation, curvature, connectivity, and volumetric envelopes as explicit spatial constraints for generative models.	Algorithmic × VR; Immersive × VR	Express three-dimensional topology and envelope constraints for protein design. Interaction consequence: Changes generative constraints, model inputs, and generated/refined outputs	Computational and experimental tests across topologies, geometry, expression, and binding Limitations: Preprint; no usability/adoption study; fidelity/designability tradeoffs; benefit boundary: Design feasibility and experimental expression	Preprint PDF pp. 2-6; Abstract and ProteinSketch system; Preprint PDF pp. 3-6; Bimanual VR platform and Fig. 1 Preprint status must remain explicit; biological feasibility is not a user-performance evaluation. | Immersive placement rests on VR-native spatial constraint specification; no learned user-preference mechanism is shown.
S059 Rapid QC-MS [57]	Rapid QC-MS: Interactive Dashboard for Synchronous Mass Spectrometry Data Acquisition Quality Control. Tasks: Algorithmic × Desktop/Planar: CFS	Each new file is automatically processed, classified against QC criteria, and shown in a synchronous interactive dashboard.	Algorithmic × Desktop/Planar	Researcher defines QC criteria; software processes each new file and classifies status; researcher diagnoses drift and decides whether to intervene.	Operational demonstrations and instrument benchmarking; no controlled user study.	Introduction ¶5; Design and workflow ¶1–3; Rapid Assessment ¶1–3
S060 RATS VR planning system [73]	Developing a Virtual Reality Simulation System for Preoperative Planning of Robotic-Assisted Thoracic Surgery. Tasks: Algorithmic × VR: NCFS	Patient CT reconstructions are segmented and rendered as manipulable VR anatomy for relation checking and resection planning.	Algorithmic × VR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X004 2. Materials and Methods ¶5

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S061 REDAC [15]	REDAC: RNA-seq expression data analysis chatbot. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	A chatbot invokes RNA-seq expression analysis and returns plots and statistical results for iterative inspection.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X039 Abstract 1 ¶2; X039 3 REDAC architecture ¶1
S062 RetainVis [36]	Large claims datasets for heart failure and cataract; Inspect and steer longitudinal clinical-risk predictions. Tasks: Algorithmic × Desktop/Planar: CFS	An interpretable recurrent model produces risk estimates and feature contributions that linked views expose for cohort and patient-history exploration and user-directed what-if steering.	Algorithmic × Desktop/Planar	Inspect and steer longitudinal clinical-risk predictions. Interaction consequence: Changes input records, contributions/model state, and predictions	Predictive comparisons, expert feedback, and fictitious case study Limitations: Authors state user problem-solving benefit was not empirically tested; benefit boundary: Predictive performance comparisons	Author report PDF pp. 1-3 and 7-9 Not placed as Adaptive: reported changes follow explicit edits and retraining commands, without evidenced system-selected tailoring; user benefit was not empirically tested.
S063 RNA-SeqEZPZ [71]	Bulk RNA-seq datasets; Run RNA-seq analysis from FASTQ through differential expression and pathways. Tasks: Algorithmic × Desktop/Planar: CFS	A point-and-click interface launches a fixed RNA-seq workflow for quality control, alignment, counting, differential expression, and pathway analysis, then exposes interactive result views.	Algorithmic × Desktop/Planar	Run RNA-seq analysis from FASTQ through differential expression and pathways. Interaction consequence: Changes run parameters and presentation; not learned model state	Public/new datasets and output comparisons Limitations: No controlled user, decision-quality, adoption, or maintenance study; benefit boundary: Technical operation and output comparison	Findings; Methods - Workflow overview and Interactive visualization Commanded pipeline execution, not Adaptive assistance; evaluation does not establish improved scientific decisions.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S064 Robin [10]	Robin: an advanced tool for comparative loop caller result analysis leveraging large language models. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	Loop-caller outputs are computed, compared, and explained through an LLM-mediated web workflow.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X018 Results ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S065 SAMIRA [63]	Towards user-centered interactive medical image segmentation in VR with an assistive AI agent. Tasks: Algorithmic × VR: NFS; Adaptive × VR: NFS; Conversational × VR: NFS; Immersive × VR: NFS	Foundation-model segmentation, retrieval-augmented guidance, and 3D rendering execute operations requested by the user. | Guidance is conditioned on the currently viewed slice, and user point prompts revise masks and propagate corrections. | Natural-language dialogue is interpreted into localization, explanation, segmentation, refinement, and visualization operations. | Head, gaze, and controller pointing specify corrections in a true-scale VR workspace while the agent updates the segmentation and guidance.	Algorithmic × VR; Adaptive × VR; Conversational × VR; Immersive × VR	The user asks, points, and corrects; the agent localizes, segments, explains, and renders; the user approves and interprets the final 3D result.	Usability evaluation and controller/head/eye input comparison retained; clinical effectiveness not established. Participant details and outcomes not established in the retained assessment.	arXiv v2 pp. 1–8; Abstract, architecture, and segmentation workflow; arXiv v2 pp. 3–10; agent workflow and refinement methods; arXiv v2 pp. 1–7; Abstract and conversational-agent design; arXiv v2 pp. 1–4, 8–14; interaction design and user study Implemented; usability is evaluated, not clinical effectiveness. | Adaptation rests on context and corrective feedback, not simple parameter entry. | Dialogue mediates analytic subgoals rather than button-equivalent commands. | Spatial input is computationally coupled to segmentation and guidance; HMD display alone is not the evidence.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S066 scSelector [25]	scSelector: A Flexible Single-Cell Data Analysis Assistant for Biomedical Researchers. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	Natural-language requests invoke single-cell filtering, comparison, and visualization operations whose results remain available for researcher inspection.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	PBMC case analysis compares seven rare cells with a control cluster using differential expression and enrichment; no controlled user benefit inferred.	X020 2. Materials and Methods > 2.1. Software Development > Main Framework Development ¶1; X020 1. Introduction ¶4; X020 1. Introduction ¶4
S067 SeqPlots [68]	SeqPlots - Interactive software for exploratory data analyses, pattern discovery and visualization in genomics. Tasks: Algorithmic × Desktop/Planar: CFS	Server-side aggregation generates heatmaps and profiles that remain linked to interactive genomic filters.	Algorithmic × Desktop/Planar	Analyst chooses regions, groups, and filters; system aggregates aligned genomic signals; analyst compares patterns.	Software validation and worked use cases; no participant study.	Abstract; Methods; Implementation; Use cases

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S068 Skin-lesion AR [24]	An Augmented Reality Mobile Application for Skin Lesion Data Visualization. Tasks: Algorithmic × AR/MR: CS; Immersive × AR/MR: CS	Image preprocessing, lesion segmentation, feature extraction, photometric stereo, and CNN classification produce parameters and a melanoma classification shown in a mobile AR view. | Spatially grounded assistance detects device-to-skin distance, centers and selects a lesion, adjusts illumination, computes a classification, and overlays the result in the dermatologist's view.	Algorithmic × AR/MR; Immersive × AR/MR	Specific analyst controls were not established in the retained assessment.	Retained placement qualification reports CNN accuracy 74.6% and a preliminary evaluation with seven dermatologists. No clinical-effectiveness result established; design and further outcomes not established in the retained assessment.	Published PDF pp. 2-5; Sections III and IV; Published PDF p. 5; Section IV Augmented Reality Visualization No repository canonical ID was found; CNN accuracy was 74.6% and the preliminary evaluation involved seven dermatologists. | Immersive placement rests on tracked spatial context and in-situ overlay, not on phone display alone; no clinical-effectiveness result is established. | Canonical report ID not recorded in the historical ledger; identity is retained through persisted group, exact title and bibliography/DOI. No canonical ID invented.

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S069 Space-Time Hypercube [23]	Immersive and interactive visualization of 3D spatio-temporal data using a space time hypercube: Application to cell division and morphogenesis analysis. Tasks: Algorithmic × VR: NCFS; Immersive × VR: NCFS	The STH projection, clipping, filtering, lineage propagation, and coordinated views transform embryo time-series data in VR. | A controller-resolved spatial cell selection drives propagation through time and lineage and synchronizes both VR representations.	Algorithmic × VR; Immersive × VR	Analyst positions cuts and selects cells; the system propagates state through lineage/time and linked views; analyst detects temporal events.	Five biologists used two one-hour sessions; wave detection was useful, but other tasks and workflows were mixed.	§3.4.1 Fig. 3; §3.4.2 ¶2–4; §3.4.3 ¶1–4; §3.4.2 ¶4; §3.4.3 ¶1–4 The counted mechanism is spatial selection feeding a computed propagation, not VR display alone.
S070 SpinalSAM-R1 [41]	SpinalSAM-R1: A Vision-Language Multimodal Interactive System for Spine CT Segmentation. Tasks: Algorithmic × Desktop/Planar: CFS; Conversational × Desktop/Planar: CFS	Vision-language prompts condition spine CT segmentation and return inspectable masks in an interactive planar workflow.	Algorithmic × Desktop/Planar; Conversational × Desktop/Planar	Specific analyst controls were not established in the retained assessment.	Retained passage reports evaluation on 120 lumbar CT scans (31,454 slices); baseline metrics and human-analytic outcomes not established in the retained assessment.	X034 PDF p. 2
S071 Surgical Theater XR platform [67, 62]	Extended reality platform for minimally invasive endoscopic evacuation of deep-seated intracerebral hemorrhage: illustrative case. | Novel planning pipeline utilizing the Surgical Theater system for pediatric epilepsy surgery. Tasks: Algorithmic × VR: NCFS; Algorithmic × AR/MR: NCFS	Segmented patient anatomy and tractography are assembled into interactive 360-degree models for VR planning and AR intraoperative reference.	Algorithmic × VR; Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	One retained clinical case describes patient-specific XR-supported planning; no comparative causal benefit inferred or transferred to all companion reports.	X057 Discussion > Observations ¶2; X057 Illustrative Case ¶2; X057 Discussion > Observations ¶2; X137 VR IN MEDICINE ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S072 Toxygates [50]	Interactive Toxicogenomics: Gene set discovery, clustering and analysis in Toxygates. Tasks: Algorithmic × Desktop/Planar: CFS	Selected gene sets are clustered, displayed as heatmaps, and passed to enrichment routines for mechanistic toxicogenomics analysis.	Algorithmic × Desktop/Planar	Analyst defines gene sets and cutoffs; system clusters and computes enrichment; analyst builds a mechanistic interpretation.	Mechanistic case study; no controlled user evaluation; dataset-normalization limits are acknowledged.	Results > Clustering ¶1; Gene set synchronisation ¶1; Case study ¶1
S073 VIADS [31]	A visual interactive analytic tool for filtering and summarizing large health data sets coded with hierarchical terminologies (VIADS). Tasks: Algorithmic × Desktop/Planar: NCFS	VIADS aggregates hierarchical terms and computes filtered summaries/comparisons with interactive preview and threshold revision.	Algorithmic × Desktop/Planar	Analyst selects algorithm and thresholds; software aggregates and compares hierarchies; analyst inspects summaries and revises thresholds.	Implementation report and algorithm examples; usability and utility studies were future work.	Background ¶2; Implementation > Design modules ¶3; Results > Algorithms ¶1; Analytic workflow ¶2
S074 VIPER [66]	Patient-specific MRI/atlas integration for functional neurosurgery. Tasks: Algorithmic × Desktop/Planar: NCFS	Automated atlas registration aligns anatomical structures to patient MRI; linked planar/3D views support inspection.	Algorithmic × Desktop/Planar	Scientist inspects registered anatomy, proposed lesions and surrounding structures. Registration editing/undo controls not established in the retained assessment.	Evaluation evidence was not established in the retained assessment.	STJEAN_1998_ atlas_integration_ neurosurgery.pdf, pp. 672–678 Desktop/Planar includes its stereoscopic monitor; stereo display alone is not Immersive assistance.
S075 Virtual embolization [65]	Virtual embolization for treatment support of intracranial AVMs using an interactive desktop and VR application. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × VR: NCFS	Precomputed embolization and flow simulations are selected and compared in a desktop application. | The VR application exposes the same simulated occlusion alternatives for spatial comparison.	Algorithmic × Desktop/Planar; Algorithmic × VR	Clinician chooses embolization scenarios; simulation produces candidate flow outcomes; clinician compares consequences.	Two neuroradiologists provided formative feedback; they valued virtual blocking but requested quantitative information.	Abstract; Results > Interactive application ¶1–4; VR exploration ¶1; Evaluation ¶1–5

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S076 Virtual Island [6]	Bringing the Field into the Lab: Large-Scale Visualization of Animal Movement Trajectories within a Virtual Island. Tasks: Algorithmic × CAVE: NCFS	Parallel computation filters individuals and time spans while integrated terrain, canopy, and GPS trajectories are explored in CAVE2.	Algorithmic × CAVE	The researcher chooses animals and time spans; parallel filtering executes; the researcher navigates the CAVE2 scene and interprets behavior in environmental context.	Evaluation evidence was not established in the retained assessment.	Author PDF pp. 1–2; Abstract, system, and interaction description Short conference report demonstrates capability, not comparative analytical benefit.
S077 Vitessce Link [47]	Vitessce Link: A Mixed Reality and 2D Display Hybrid Approach for Visual Analysis of 3D Tissue Maps. Tasks: Algorithmic × Desktop/Planar: NCFS; Algorithmic × AR/MR: NCFSR; Immersive × AR/MR: CFS	Linked selections and multiscale tissue-map computations coordinate a planar Vitessce view with registered MR representations. | Tracked finger positions specify measurement endpoints; geometric distance is computed within the tissue map.	Algorithmic × Desktop/Planar; Algorithmic × AR/MR; Immersive × AR/MR	Scientist specifies hand-position endpoints for tissue measurement and inspects linked planar/MR views. Detailed undo and intervention controls not established in the retained assessment.	Kidney and melanoma case studies and user/community feedback are retained; these do not independently establish improved analytical performance.	X015 PDF p. 11; X015 PDF p. 2; X015 PDF p. 7 | Measurement-based Immersive pairing only; synchronization, navigation and collaboration alone are not the basis. No Adaptive or Conversational addition or independently validated performance gain.
S078 VROOM [37]	Virtual reality for the observation of oncology models (VROOM): immersive analytics for oncology patient cohorts. Tasks: Algorithmic × VR: NCFS	Patient-cohort variables and oncology models are computationally arranged for immersive VR comparison and filtering.	Algorithmic × VR	Specific analyst controls were not established in the retained assessment.	A case-study walkthrough is mentioned in the retained conclusion; evaluation design and findings not established in the retained assessment.	X032 Results and discussions > VROOM case study: AML ¶3; X032 Abstract 1 ¶1

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System / reports	Application / tasks	Mechanism	Supported pairs	Scientist role	Evaluation / limits	Evidence / qualifications
S079 VR–AR medical-image analysis workflow [53]	Enhanced Medical Image Analysis: Leveraging 3D Visualization and VR-AR Technology. Tasks: Algorithmic × VR: NCFS; Algorithmic × AR/MR: NCFS	Implemented preprocessing, volume/surface reconstruction, rendering, slicing, and gesture interaction transform medical images for VR/AR inspection.	Algorithmic × VR; Algorithmic × AR/MR	Specific analyst controls were not established in the retained assessment.	Assessment unresolved: retained evaluation field contains background, architecture, or capability wording rather than a sufficiently attributable evaluation finding.	X139 PDF pp. 3-7, Proposed Model, Experiment, and Result Capability is reported in a preprint; participant and model-evaluation reporting is limited and not independently replicated.

8.1 Eligible Systems Without Recorded Supported Placements

Table S5 completes the eligible-system inventory using the retained report–system crosswalk. It reports existing assessment states, not new exclusions or placements. Failure to support a targeted combination does not resolve other assistance–modality combinations; none is added retrospectively here.

Table S5: Four eligible systems without a recorded supported pairing. R identifiers refer to the selection-flow crosswalk; TSC identifiers refer to the targeted-coverage matrix.

System and trace	Report identity	Retained placement boundary
CAVEman (R038; Q18)	*CAVEman: Standardized anatomical context for biomedical data mapping*; `https://doi.org/10.1002/ASE.3`	A related report supports semantic and visual mapping, but describes only a virtual-reality environment; the queued primary report and CAVE hardware were unverified. No pairing was recorded.
LiverPlan (R319; T049; TSC-055)	*LiverPlan: A Stage-Adaptive Immersive Visual Analytics Framework for Anatomical Liver Surgical Planning*; canonical record `0be187f0ef0f3a0fb8c87105`	The report's implemented mechanisms operate in a headset-based XR prototype. Desktop was an evaluation baseline, so the targeted Immersive–Desktop/Planar pairing was unsupported. Other placements were not established by this targeted assessment.
YASARA Model VR (R321; T054; TSC-061)	*YASARA Model—Interactive Molecular Modeling from Two Dimensions to Virtual Realities*; `https://doi.org/10.1021/acs.jcim.3c01136`	Desk apparatus is represented inside VR; this did not support the targeted Immersive–Desktop/Planar pairing. Other pairings are not inferred from the rejected target.
Semantic molecular-analysis pipeline (R338; T077; TSC-087)	*Semantics for an Integrative and Immersive Pipeline Combining Visualization and Analysis of Molecular Data*; `https://doi.org/10.1515/jib-2018-0004`	The proof of concept concerns immersive/CAVE, AR, and screen-wall contexts. The targeted Immersive–Desktop/Planar mechanism was not demonstrated; other pairings were not established in this assessment.

The crosswalk retains automated assessment authority for these records; this inventory supplies no new author approval or independent validation. CAVEman's qualification is also recorded in `evidence/candidate-census/cumulative_eligibility_assessments.csv`; the other three target decisions are retained in `evidence/targeted-coverage/deduplicated_target_system_cell_matrix.csv`.

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